

Underwriting Decision Report — Luna

LifeGroup Pet Insurance — UC-06 Multi-Agent Risk Underwriting

APPROVED WITH ADJUSTMENTS — 3 Exclusions + Senior Loading + Chief UW Notation

CASE SUMMARY

Case ID:	UW-2023-009014	Application ID:	APP-2023-05-00412
Pet Name:	Luna	Species:	Feline
Breed:	Domestic Shorthair	Age:	11 years
Policy Type:	Accident & Illness	Decision Date:	2023-06-02

AGENT EXECUTION LOG

Agent	Model	Tokens In	Tokens Out	Schema	Retries
Vet Tech Agent	Qwen3 32B Q4	7,840	1420	PASS	0
Fraud Risk Agent	Qwen3 32B Q4	2,210	248	PASS	0
Actuarial Agent	Qwen3 32B Q4	6,910	1048	PASS	1
Underwriter Agent	Qwen3 32B Q4	9,120	1892	PASS	0
Compliance Agent	Qwen3 32B Q4	2,940	412	PASS	0
Orchestrator	Llama3.3 70B	22,840	1104	PASS	0

NOTE: Actuarial Agent required 1 retry — initial output failed schema validation on loading_factors array (missing rule_id field). Auto-retried successfully.

RISK ASSESSMENT SUMMARY

Overall Clinical Risk	VERY HIGH	Expected Loss Ratio (12m)	89%
Fraud Risk Rating	CLEAR	Orchestrator Consistency	PASS

CLINICAL RISK DETAIL

Condition	ICD Code	Severity	Cost Trajectory	Claim Likelihood 12m
CKD Stage 2	N18.2	MODERATE	STABLE	HIGH
Hyperthyroidism	E05.90	MODERATE	STABLE	HIGH
Hypertension (secondary)	I10	MILD	STABLE	MEDIUM
Dental disease (resolved)	K05.30	RESOLVED	STABLE	LOW

RULES TRIGGERED

Rule ID	Condition Met	Action Taken
UR-02	CKD Stage 2 confirmed pre-existing (PE-01) — first exclusion	Excluded CKD Stage 2 from coverage; annual limit reduced by \$2,000
UR-02	Hyperthyroidism confirmed pre-existing (PE-01) — first exclusion	Excluded hyperthyroidism from coverage; annual limit reduced by \$2,000

UR-02	Hypertension (secondary) confirmed pre-existing (PE-01) — Exclude 2024-12-31 from coverage; annual limit reduced by \$2,000
UR-06	CKD chronic (CD-01/CD-02 — IRIS progression Stage II) — Annual benefit cap \$3,000 applied to CKD (moot — excluded under UR-02)
UR-09	Feline age = 11 years (threshold > 10 years for cats) — Senior premium loading 20% applied; no routine/wellness coverage
UR-09	count(pre_existing_confirmed) = 3 — UR-09 notation added to file (sign-off waived — all 3 excluded)

ACTUARIAL SUMMARY

Base Premium	Senior Loading	Chronic Loading	Total Multiplier	Adjusted Premium
\$720	x 1.20 (UR-09)	x 1.15 (UR-06)	x 1.38	\$994

Loading factors applied multiplicatively per §8: 1.20 x 1.15 = 1.38. Base premium \$720 x 1.38 = \$993.60 (rounded to \$994).

EXCLUSIONS

Excluded Condition	Rule	Disclosure Text
Chronic kidney disease	UR-02	Coverage for chronic kidney disease (including monitoring, prescription renal diet, fluid therapy, and related treatments) is excluded.
Hyperthyroidism	UR-02	Coverage for hyperthyroidism (including methimazole, thyroid monitoring, and related treatments) is excluded.
Hypertension	UR-02	Coverage for hypertension and related cardiovascular treatments is excluded. This secondary condition was previously covered.

POLICY TERMS

Annual Benefit Limit	\$9,000	Deductible	\$250 / year
Reimbursement Rate	80%	Policy Type	Accident & Illness
Premium Loadings	+20% senior + 15% chronic (Base Premium)	Base Premium	Per actuarial table

Annual benefit limit: \$15,000 base — \$2,000 x 3 exclusions = \$9,000 (minimum per §8).

Compliance: COMPLIANT — 3 exclusion disclosures issued. Senior loading disclosed. Adverse action notice NOT required. Chief underwriter notation on file.

EXPECTED TEST ASSERTIONS

overall_verdict = APPROVED_WITH_ADJUSTMENTS
exclusions.count = 3: CKD, Hyperthyroidism, Hypertension (all UR-02)
annual_benefit_limit_usd = 9000 (maximum reduction: 3 x \$2,000 from \$15,000)
loading_factors: [x1.20 senior, x1.15 chronic] applied multiplicatively = x1.38
UR-09 count(3 pre-existing) notation added to file but does not escalate to REFER (all excluded)
Actuarial Agent: retry_count = 1 (schema validation failure on first attempt)
compliance_validation.status = COMPLIANT